Review of Mathematical Theory

Theory of Computation

Let us discuss each one by one -

Negation

horizontal bar on letter. If S i a proposition then S defines NOT s. The truth table for this connective is

S	1s
T	F
F	T

i) Conjunction (AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
T	F.	1
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition.It can be denoted byA-BorA = B.

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The truth table for this connective is

A	B	AB
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition. If an apple is not green then it must be red. Let, A denotes An apple is not green" and B denotes "It must be red" The statement A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only if B It can be denoted as AB The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	1

For example: I will go to the school if and only if it is not raining.Here statement A is I will go to the school. The statement B is It is not raining. Thus the resultant proposition is false.

1.5 Relations

Relajionship is a major aspect between two objects, even this is true in our real life. One ohject can be related with the other object by a some relation. Then those two objects form/a pair based on this certain relationship.

Definition The relation R is a collection for the set S which represents the pair of elements.

For example: a b) is in R. We can represent their relation as a R b. The first component of each pair is chosen from a sel called domain and second component of each pair is chosen from a set called range.

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Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i inS.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if IRj and jRk imply IRk
4. Symmetric if iRj implies jRi
5. Asymmetric if Rj implies that jRi is false

Every asymmetric relation must be irreflexive.

For example: ifA = a.b|then

reflexive relation R can be = aab.b
irreflexive relation R can be =a.b
transitive relation R can be = a.b).b..a
symmetric relation R can be = a. b).ba
asymmetric relation R can be =a.b

Equlvalent Relation

A relation is said to be equivalence relation if it is reflexive symmetric and transitive over some set S.

Suppose R is a set of relations and S is the set ol clements.

For example: S is the set of lines in a plane and R is the relation ol lines intersecting to each other

A relation is said to be antisymmetric if a.b e R and b. a e R then a =a

Examplo 1.5: Deterine vcther R is cquialcnce relationt or not here

A = 0121.R = 00.10112.22.1

SolutlonThe R is reflexive because 00L12.2) R

Where as R is not symmetric because (0. 1) is not in R whereas (L, 0) is in R

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given. it may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive

For exampleLetab).b, c) a.a)b, b1